

Reader Universal Function Library
V2.2

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1. Instruction

The development kit supports the ANSI-C, ANSI-C++ and other languages which are compatible with C Language interface, such as delphi, VB, C#. It supports 32-bit/64-bit Windows operating systems (Windows XP, Windows 7, Windows 8, and Windows 10).The development kit cannot be used independently and must be combined with other RFID development kit. Before development, in order to know the whole process of SDK, you must read “DLL Development Guide.doc”first.

The "rfidlib-reader" development package is a common API interface congregation in our SDK, which contains the API to tag inventory, tag connect, tag access, tag disconnect, interface for device and security door operations and other control commands and so on.

2. Application Integration

This package consists of the following files :

Files	Description
rfidlib_reader.dll	Dynamic library, contains all API interface
rfidlib_reader.lib	Used for VC++ project
rfidlib_reader.h	Header file of C/C++ project, the API definition
rfidlib_reader.cs	C# class file
rfidlib_reader.pas	Header file of Delphi definition

Programming Language	Description
VC++	Copy rfidlib_reader.lib and rfidlib_reader.h to the project directory and add to the project. Copy rfidlib_reader.dll to the exe file output directory.
C#	Copy rfidlib_reader.cs to the project directory and add to the project. Copy rfidlib_reader.dll to the exe file output directory.
Delphi	Copy rfidlib_reader.pas to the project directory and add to the project. Copy rfidlib_reader.dll to the exe file output directory.

3. Programming Interface

3.1. RDR_LoadReaderDrivers

Definition	err_t RDR_LoadReaderDrivers(LPCTSTR path)			
Description	Load all of the reader driver. This API should be immediately used after the application finish initialization.			
Parameter	Name	Type	Direction	Remark
	path	LPCTSTR	[in]	1. If it is an empty string, then it will point to the path which is follow "rfidlib_reader.dll". 2. If it start with '\', it will point to the subdirectory of 'rfidlib_reader.dll' position. 3. The path is an absolute address.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.2. RDR_Open

Definition	err_t RDR_Open(LPCTSTR connStr ,RFID_READER_HANDLE *hrOut)			
Description	Open the device driver specified by the connection string			
Parameter	Name	Type	Direction	Remark
	connStr	LPCTSTR	[in]	Device connection string, About the connection string description, please refer to “DLL Development Guide.doc”.
	hrOut	RFID_READER_HANDLE *	[out]	Return devices handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.3. RDR_Close

Definition	err_t RDR_Close(RFID_READER_HANDLE hr)			
Description	Close devices drivers			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Devices drivers handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.4. HID_Enum

Definition	DWORD HID_Enum(LPCTSTR DevId)			
Description	Enumerate all USBHID devices of this type. DevId include reader name or model code			
Parameter	Name	Type	Direction	Remark
	DevId	LPCTSTR	[in]	Reader name or model code
Return	Number of devices enumerated			
Reference Code				

3.5. HID_GetEnumItem

Definition	err_t HID_GetEnumItem(DWORD indx ,BYTE infType,LPTSTR infBuffer,DWORD *nSize)			
Description	Obtain information about the devices enumerated			
Parameter	Name	Type	Direction	Remark
	indx	DWORD	[in]	The index number of the buffer array which enumerated by HID_Enum function.The value ranges from 0 to HID_Enum API return value - 1
	infType	BYTE	[in]	The type of information to obtain: 1: Device serial number; 2: The path where the device driver exists in the operating system;
	infBuffer	LPTSTR	[out]	Byte buffer, used to store information.
	nSize	DWORD *	[in,out]	In: The maximum number of bytes in the buffer; Out: The number of bytes written to the buffer;
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.6. COMPort_Enum

Definition	DWORD COMPort_Enum(void)			
Description	Enumerate the serial ports of the Windows system			
Parameter	Name	Type	Direction	Remark
Return	Number of serial ports enumerated			
Reference Code				

3.7. COMPort_GetEnumItem

Definition	<code>err_t COMPort_GetEnumItem(DWORD idx,LPTSTR nameBuf,DWORD nSize)</code>			
Description	Obtain the enumerated serial port names, such as “COM1”			
Parameter	Name	Type	Direction	Remark
	idx	DWORD	[in]	Index number of the buffer array enumerated by the COM_Enum function. The value ranges from 0 to COM_Enum Return quantity -1.
	nameBuf	LPTSTR	[out]	Byte buffer, used to save information.
	nSize	DWORD	[in]	The maximum number of bytes that can be written to the buffer;
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.8. Bluetooth_Enum

Definition	DWORD RFIDLIB_API Bluetooth_Enum()			
Description	Enumerate already matching bluetooth devices			
Parameter	Name	Type	Direction	Remark
Return	Return the number of that already matched bluetooth			
Reference Code				

3.9. Bluetooth_GetEnumItem

Definition	err_t RFIDLIB_API Bluetooth_GetEnumItem(DWORD idx ,BYTE infType,LPTSTR nameBuf,DWORD *nSize)			
Description	Get information of enumeration bluetooth			
Parameter	Name	Type	Direction	Remark
	idx	DWORD	[in]	Index, starting from 0
	infType	BYTE	[in]	Type 1: get the bluetooth name 2: get the bluetooth address
	nameBuf	LPTSTR	[in]	Buffer used of saving the bluetooth information
	nSize	DWORD*	[in]、 [out]	[in]: the size of buffer [out]: data actual length that saved in buffer
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.10.DNODE_Destroy

Definition	err_t DNODE_Destroy(RFID_DN_HANDLE dn)			
Description	Release the created data node, such as RDR_CreateInvenParamSpecList , RDR_CreateSetOutputOperations .			
Parameter	Name	Type	Direction	Remark
	dn	RFID_DN_HANDLE	[in]	Handle of data node
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.11.RDR_CreateInvenParamSpecList

Definition	RFID_DN_HANDLE RDR_CreateInvenParamSpecList() ;
Description	Create an air interface protocol parameter list data node, if not use, need to call DNODE_Destroy to release resources
Parameter	None
Return	If successful, returns the handle of data node.
Reference Code	

3.12.RDR_TagInventory

Definition	err_t RDR_TagInventory(RFID_READER_HANDLE hr, BYTE AIType, BYTE AntennaCount, BYTE AntennaIDs[] , RFID_DN_HANDLE InvenParamSpecList) ;				
Description	Inventory the tags. This API supports multiple antenna interfaces and multiple air interface protocols. The tag which was be finded, willll be save as mode of data node format. After inventory, the application should use the API ‘RDR_GetTagDataReport’ to get all tag’s information.				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	AIType	BYTE	[in]	Query Type	
				Value	Description
				1	New query. Before Inventory, all tags are in ready status;
				2	Continue query. The tags cannot be read if they are in quiet status;
	AntennaCount	BYTE	[in]	The number of antennas used for inventory. If 0, the reader will use the default antenna configuration.	
	AntennaIDs	BYTE []	[in]	The list of the antennas. When AntennaCount=0, this value is NULL.	
	InvenParamSpecList	RFID_DN_HANDLE	[in]	Air interface protocol parameter list (Include one or more air interface protocol parameters). If the list contains both ISO15693 and ISO14443A protocols, the reader should read these two types of tags (The reader must support these protocols).	
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				

Reference Code	<pre> err_t iret ; BYTE AIType ; RFID_DN_HANDLE dnhInvenParamSpecList; dnhInvenParamSpecList = RDR_CreateInvenParamSpecList(); /* ISO15693 protocol tags*/ ISO15693_CreateInvenParam(dnhInvenParamSpecList,0,FALSE ,0,0) ; /* Add other protocol */ /*start to find tags */ AIType = 1 ; //new search LABEL_DOINVENTORY: iret=RDR_TagInventory (hr, AIType,0,NULL, dnhInvenParamSpecList) ; if((iret == 0) (iret == -ERR_STOPTRRIGOCUR/* The search process meet the exit criteria */)) { /* Parsing tag data report*/ DWORD nTagCount ; nTagCount = RDR_GetTagDataReportCount (TagDataReports); // Get the first report RFID_DN_HANDLE dnhTagReport = RDR_GetTagDataReport(hr,RFID_SEEK_FIRST) ; While(dnhTagReport) { UINT APItype,TagType,AntId ; BYTE dsfid; BYTE uid[8]; iret = ISO15693_ParseTagDataReport(dnhTagReport,& APItype , & TagType , &AntId,&dsfid,uid) ; if(iret == NO_ERR) { //Is ISO15693 tag } //Next report dnhTagReport = RDR_GetTagDataReport(hr,RFID_SEEK_NEXT) ; } if(iret == -ERR_STOPTRRIGOCUR) { AIType= 2; //Get the rest of the tags goto LABEL_DOINVENTORY ; } } If(dnhInvenParamSpecList) DNode_destroy(dnhInvenParamSpecList) ; </pre>
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3.13.RDR_GetTagDataReportCount

Definition	DWORD RDR_GetTagDataReportCount(RFID_READER_HANDLE hr)			
Description	Get the tag data report quantity of inventory			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDL E	[in]	Reader handle
Return	Returns the count of tags which found in the inventory			
Reference Code				

3.14.RDR_GetTagDataReport

Definition	RFID_DN_HANDLE RDR_GetTagDataReport(RFID_READER_HANDLE hr , BYTE seek)				
Description	Get data node of tag data report in inventory				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	seek	BYTE	[in]	The tag data report position	
				Value	Description
				0	The cursor position remains unchanged, get current record.
				1	The cursor move to first, get current record
				2	The cursor move to next, get current record
				3	The cursor move to last, get current record
Return	Return data node of tag data report that specified by current cursor.				
Reference Code					

3.15.RDR_TagDisconnect

Definition	err_t RDR_TagDisconnect(RFID_TAG_HANDLE hTag)			
Description	Disconnect with tag and release resource of tag handle			
Parameter	Name	Type	Direction	Remark
	hTag	RFID_TAG_HANDLE	[in]	Tag handle of already connection
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.16.RDR_GetReaderLastReturnError

Definition	err_t RFIDLIB_API RDR_GetReaderLastReturnError(RFID_READER_HANDLE hr)			
Description	If specific operation is failure and API return -17 error, it means the hardware is incorrect. The application can get the specific error type via this API.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	>=0 is error code that says reader hardware return operation failed			
Reference Code				

3.17.RDR_DisconnectAllTags

Definition	err_t RDR_DisconnectAllTags(RFID_READER_HANDLE hr)			
Description	Disconnect all of the tags and release resource of all the tag handle.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.18.RDR_SetAccessAntenna

Definition	err_t RDR_SetAccessAntenna(RFID_READER_HANDLE hr ,BYTE AntennaID)			
Description	For the multiple antennas reader, if the tag need to be read/write the tag, can through this API set the range of antenna sensing.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	AntennaID	BYTE	[in]	Antenna ID
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.19.RDR_OpenRFTransmitter

Definition	err_t RDR_OpenRFTransmitter(RFID_READER_HANDLE hr)			
Description	Open RF output			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.20.RDR_CloseRFTransmitter

Definition	err_t RDR_CloseRFTransmitter(RFID_READER_HANDLE hr)			
Description	Close RF transmitter			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	0: Success			
	Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.21.RDR_SetCommuImmeTimeout

Definition	err_t RDR_SetCommuImmeTimeout(RFID_READER_HANDLE hr)			
Description	make the API timeout that communicates with the reader to exit			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.22.RDR_GetAntennaInterfaceCount

Definition	DWORD RDR_GetAntennaInterfaceCount(RFID_READER_HANDLE hr) ;			
Description	Get the quantity of antenna interface that reader support			
Parameter	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	The count of antenna interface			
Reference Code				

3.23.RDR_GetOutputCount

Definition	err_t RDR_GetOutputCount(RFID_READER_HANDLE hr,BYTE *nCount) ;			
Description	Get quantity of output port that support by reader			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	nCount	BYTE *	[out]	Return quantity of output port that support by reader
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.24.RDR_GetOutputName

Definition	err_t RDR_GetOutputName(RFID_READER_HANDLE hr,BYTE idxOut,LPTSTR bufName ,DWORD *nSize)			
Description	Get the name of output port			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	idxOut	BYTE *	[in]	Index of output port, start with 1
	bufName	LPTSTR	[out]	The buffer used for save the port name.
	nSize	DWORD*	[in,out]	In: the max size of the buffer. Out: the actually size written to buffer.
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.25.RDR_CreateSetOutputOperations

Definition	RFID_DN_HANDLE RFIDLIB_API RDR_CreateSetOutputOperations() ;			
Description	Create operation list parameter of output port which use in 'RDR_SetOutput'. If the parameter is not in use any more, the application must use 'DNode_Destroy' to release the memory.			
Parameter	Name	Type	Direction	Remark
Return	Handle of operation list parameter in output port			
Reference Code				

3.26.RDR_AddOneOutputOperation

Definition	err_t RDR_AddOneOutputOperation(RFID_DN_HANDLE hOperas,BYTE outNo,BYTE outMode,DWORD outFrequency,DWORD outActiveDuration ,DWORD outInactiveDuration) ;				
Description	Add an IO port operation to the operation handle.				
Parameter	Name	Type	Direction	Remark	
	hOperas	RFID_DN_HANDLE	[in]	Handle of IO port operation,created by RDR_CreateSetOutputOperations	
	outNo	BYTE	[in]	Index of output port, start with 1	
	outMode	BYTE	[in]	Output mode	
				Value	Description
				0	Use the default configuration of reader
				1	Always open
				2	Always close
	3	Output port flashing			
	outFrequency	DWORD	[in]	Flashing frequency, only effective when outMode =3	
outActiveDuration	DWORD	[in]	Output port activation time , only effective when outMode =3		
outInactiveDuration	DWORD	[in]	Output port pause time, only effective when outMode =3 and outFrequency>1		
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.27.RDR_SetOutput

Definition	err_t RDR_SetOutput(RFID_READER_HANDLE hr ,RFID_DN_HANDLE outputOps) ;			
Description	Set IO port output.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HAN DLE	[in]	Reader handle
	outputO pers	RFID_DN_HANDLE	[in]	Output port control operation list , created by 'RDR_CreateSetOutputOperations'
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.28.RDR_SystemReset

Definition	err_t RDR_SystemReset(RFID_READER_HANDLE hr) ;			
Description	Restart the device			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HAN DLE	[in]	Reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.29.RDR_ResetCommuImmeTimeout

Definition	err_t RDR_ResetCommuImmeTimeout(RFID_READER_HANDLE hr) ;			
Description	If app call 'RDR_SetCommuImmeTimeout' to exit thread, then after exit thread need to call this api immediately, otherwise next command operation would return error -5 probably			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Result				

3.30.RDR_GetReaderInfor

Definition	err_t RDR_GetReaderInfor(RFID_READER_HANDLE hr, BYTE Type ,LPTSTR buffer, DWORD *nSize) ;			
Description	Get information of reader			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	Type	BYTE	[in]	Type of device information. This value is 0 in the operation.
	buffer	LPTSTR	[out]	String buffers , used for saved the device information. Output formats: “firmware version; device type; serial number”.
	nSize	DWORD *	[in,out]	In: the max size of string buffer. Out: the size written to the buffer. If the size is not enough, the string will be truncated.
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.31.RDR_ConfigBlockWrite

Definition	err_t RDR_ConfigBlockWrite(RFID_READER_HANDLE hr,DWORD cfgno ,BYTE cfgdata[],DWORD nSize,DWORD mask) ;			
Description	Write data to configuration block, the configuration block definition of each card reader is different, so before call this function ,please read user drivers document part of relative reader driver			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	cfgno	DWORD	[in]	Address of configuration block that to write
	cfgdata	BYTE[]	[in]	Byte array , include configuration block data that to write, max support 32 byte
	nSize	DWORD	[in]	Byte size of configuration block data that to write
	mask	DWORD	[in]	The 32 bits of this parameter correspond to 32 bytes of configuration block data to be written, If a bit of this parameter is 1, it means that the byte corresponding to this bit needs to be updated, otherwise will not be update
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.32.RDR_ConfigBlockRead

Definition	err_t RDR_ConfigBlockRead(RFID_READER_HANDLE hr,DWORD cfgno,BYTE cfgbuff[],DWORD nSize)			
Description	Read configuration block data, the configuration block definition of each card reader is different, so before call this function ,please read user drivers document part of relative reader driver			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	cfgno	DWORD	[in]	Address of configuration block that to read
	cfgdata	BYTE[]	[out]	The bytes array used for saving the data.
	nSize	DWORD	[in]	Max length of char that can be saved in 'cfgdata' parameter
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.33.RDR_ConfigBlockSave

Definition	err_t RDR_ConfigBlockSave(RFID_READER_HANDLE hr,DWORD cfgno)			
Description	Use this API to save configuration block data and the data will not be lost after the reader restart.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	cfgno	DWORD	[in]	Address of configuration block to save
Return	0: Success Not 0: Failure, please check RFIDLJB API Error code table.			
Reference Code				

3.34.RDR_CreateRS485Node

Definition	err_t RDR_CreateRS485Node(RFID_READER_HANDLE hr ,DWORD busAddr,RFID_READER_HANDLE *ohrRS485Node)			
Description	Create RS485 node base on reader drivers			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	busAddr	DWORD	[in]	RS485 Bus address: 1-254
	ohrRS485Node	RFID_READER_HANDLE *	[out]	If create success ,export reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.35.RDR_SetEventHandler

Definition	err_t RDR_SetEventHandler(RFID_READER_HANDLE hr,BYTE eventType,BYTE methType,UINT msg ,HWND hwnd,RFID_EVENT_CALLBACK_NEW cb,VOID *param)				
Description	Set the event handler, for specific event, please check the description of device driver manual				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	eventType	BYTE	[in]	Event Type , need to check the description ‘AppendixA’.	
	methType	BYTE	[in]	Method type of event notification	
				Value	Description
				1	Use ‘SendMessage’ to notify the application layer
				2	Use callback function to notify the application layer
	3	Use ‘PostMessage’ to notify the application layer			
	msg	UINT	[in]	Windows message	
	hwnd	HWND	[in]	Windows handle	
cb	RFID_EVENT_CALLBACK_NEW	[in]	Callback function		
param	VOID *	[in]	Parameters of callback function		
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.36.RDR_GetAirInterfaceProtName

Definition	err_t RDR_GetAirInterfaceProtName(RFID_READER_HANDLE hr ,DWORD AIPType ,LPTSTR namebuf,DWORD nSize)			
Description	Get air interface type name (only supports English name)			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	AIPType	DWORD	[in]	Air interface protocol type, please check the appendix 2 air interface protocol type code of “DLL Development Guide.doc”
	namebuf	LPTSTR	[out]	The buffer used for saving the air interface name.
	nSize	DWORD	[in]	The max size that can be written in ‘namebuf’.
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.37.RDR_LoadAuthKey

Definition	err_t RDR_LoadAuthKey(RFID_READER_HANDLE hr ,BYTE storeType,BYTE keyType ,BYTE key[] ,WORD keylen)				
Description	Download the authentication key. Different devices in the key storage area may be different, need to refer Devices drivers manual.				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	storeType	BYTE	[in]	Store Type	
				Value	Description
				0	Save in RAM
	1	Save in RAM and ROM			
	keyType	BYTE	[in]	Authentication key type in storage, type descript refer to Devices drivers manual	
	key	BYTE []	[in]	byte buffer , include key data that to write	
keylen	WORD	[in]	The data length of key		
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.38.RDR_SelectAuthKey

Definition	err_t RDR_SelectAuthKey(RFID_READER_HANDLE hr, BYTE keyType) ;			
Description	Select the authentication key			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	keyType	BYTE	[in]	The authentication key types of storage area, type specification please see the instruction manual of device drivers
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.39.RDR_SetInvenStopTrigger

Definition	err_t RDR_SetInvenStopTrigger(RFID_DN_HANDLE hInvenParams,BYTE stopTriggerType,DWORD maxTimeout,DWORD triggerValue) ;				
Description	Set the stop trigger of inventory operation				
Parameter	Name	Type	Direction	Remark	
	hInvenParams	RFID_DN_HANDLE	[in]	The handle which is created by ‘RDR_CreateInvenParamSpecList’.	
	stopTriggerType	BYTE	[in]	Stop trigger type:	
				Value	Description
				0x00	When no new tags within specified time or max timeout, exit inventory process, triggerValue fields represents the time value
				0x01	If no new tags are found or the maximum timeout time expires after attempting to send N card reading commands, the inventory process will exit. The ‘triggerValue’ field represents N values.
				0x02	When N tags are found or the maximum timeout time expires, the inventory process will exit. The ‘triggerValue’ field represents N values
	0x03	The max waiting timeout , ‘triggerValue’ field remains 0.			
	maxTimeout	DWORD	[in]	The max timeout	
triggerValue	DWORD	[in]	The meaning of this value is decided by “stopTriggerType” fields		
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.40.RDR_GetSystemTime

Definition	err_t RDR_GetSystemTime(RFID_READER_HANDLE hr ,DWORD *year,BYTE *month,BYTE *day ,BYTE *hour,BYTE *minute,BYTE *second)			
Description	Get system time of device			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
	year	DWORD *	[out]	Year
	month	BYTE *	[out]	Month
	day	BYTE *	[out]	Day
	hour	BYTE *	[out]	Hour
	minute	BYTE *	[out]	Minute
	second	BYTE *	[out]	Second
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.41.RDR_SetSystemTime

Definition	err_t RDR_SetSystemTime(RFID_READER_HANDLE hr ,DWORD year,BYTE month,BYTE day ,BYTE hour,BYTE minute,BYTE second)			
Description	Set system time of device			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
	year	DWORD	[in]	Year
	month	BYTE	[in]	Month
	day	BYTE	[in]	Day
	hour	BYTE	[in]	Hour
	minute	BYTE	[in]	Minute
	second	BYTE	[in]	Second
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.42.RDR_LoadFactoryDefault

Definition	err_t RFIDLIB_API RDR_LoadFactoryDefault(RFID_READER_HANDLE hr) ;			
Description	Restore factory setting			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4. API Interface Used for Device Buffering Mode

The following API support the security door to obtain alarm records and support r-pan to obtain buffer records in scanning mode.

4.1. RDR_BuffMode_FetchRecords

Definition	err_t RDR_BuffMode_FetchRecords(RFID_READER_HANDLE hr ,DWORD flags)			
Description	In buffer mode, to get buffer record saved in devices			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	flags	DWORD	[in]	flag: Value 0: Get the record pointed to by the current record cursor; Value 1: Moves the current record cursor to the new position (current position + number of successfully uploaded records) and uploads the record of the new position.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4.2. RDR_BuffMode_ClearRecords

Definition	err_t RDR_BuffMode_ClearRecords(RFID_READER_HANDLE hr);			
Description	In buffered mode, clear buffered records			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4.3. RDR_BuffMode_FlashEmpty

Definition	err_t RDR_BuffMode_FlashEmpty(RFID_READER_HANDLE hr) ;			
Description	In buffered mode, the records stored in Flash are erased, this command can not be called frequently because the Flash lifetime is limited.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4.4. RDR_ParseTagDataReportRaw

Definition	err_t RDR_ParseTagDataReportRaw(RFID_DN_HANDLE hTagReport,BYTE rawBuffer[],DWORD *nSize) ;			
Description	Parse 'TagDataReport' as raw data output			
Parameter	Name	Type	Direction	Remark
	hTagReport	RFID_DN_HANDLE	[in]	Handle of tag data report
	rawBuffer	BYTE []	[out]	Byte buffer to store raw data.
	nSize	DWORD *	[in,out]	in: The maximum number of bytes the buffer can write; out: The number of bytes written to the buffer.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4.5. RDR_BuffMode_StartReportCollection

Definition	err_t RDR_BuffMode_StartReportCollection(RFID_READER_HANDLE hr)			
Description	Starts the acquisition of buffered recording of device in buffer mode ,The function will create a thread that automatically calls the RDR_BuffMode_FetchRecords function. An event of RFIDLIB_EVNT_GETBUF_REPORT_FOUND is generated when a record is acquired. When the function returns an error, an event of RFIDLIB_EVNT_GETBUF_ERROR is generated. If the application wants to receive an event, the RDR_SetEventHandler is invoked to set the notification method of the event			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4.6. RDR_BuffMode_StopReportCollection

Definition	err_t RDR_BuffMode_StopReportCollection(RFID_READER_HANDLE hr)			
Description	The function stops the acquisition and waits for the thread to exit completely			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

4.7. RDR_BuffMode_StopReportCollectionNoWait

Definition	err_t RDR_BuffMode_StopReportCollectionNoWait(RFID_READER_HANDLE hr)			
Description	The function sets the exit flag of acquisition thread on, but does not wait for the thread to exit completely, and the function returns immediately. This function must combine with the event RFIDLIB_EVNT_GETBUF_EXIT. If the RDR_SetEventHandler is set to return the RFIDLIB_EVNT_GETBUF_EXIT event, the event is sent to the application before the acquisition thread exits, and the application can then assume that the acquisition thread has completely exited			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5. Security door API

5.1. RDR_ResetPassingCounter

Definition	err_t RDR_ResetPassingCounter(RFID_READER_HANDLE hr , DWORD flag)			
Description	Reset passing counter			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
	flag	DWORD	[in]	Clear flag
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

flag: Clear flag

Bit	LSG405/LSG606	LSG428/LSG422/LSG618
0x01	Clear the accumulated in traffic count of this device	Clear the accumulated in traffic count of the system
0x02	Clear the accumulated out traffic count of this device	Clear the accumulated out traffic count of the system
0x03	Clear the accumulated pedestrian total traffic count of this device	Clear the accumulated pedestrian total traffic count of the system

5.2. RDR_GetPassingCounter

Definition	err_t RDR_GetPassingCounter(RFID_READER_HANDLE hr ,DWORD *inFlow,DWORD *outFlow) ;			
Description	Get passing counter			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
	inFlow	DWORD *	[out]	Return In passing counter
	outFlow	DWORD *	[out]	Return out passing counter
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

Field name	LSG405/LSG606	LSG428/LSG422/LSG618
inFlow	The accumulated in traffic count of this device	The accumulated in traffic count of the system
outFlow	The accumulated out traffic count of this device	The accumulated out traffic count of the system

5.3. RDR_ReverseInOutDirection

Definition	err_t RDR_ReverseInOutDirection(RFID_READER_HANDLE hr)			
Description	Reverse the in and out direction, The function is mainly used for library security gate and meeting gate, with infrared sensor function			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Handle of device driver
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.4. LSG_GetIRCurrentSta

Definition	err_t RFIDLIB_API LSG_GetIRCurrentSta(RFID_READER_HANDLE hr, BYTE * ir_num, BYTE * irCoverBitMask);			
Description	Obtain the infrared shielding status of the security door, only support LSG405			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ir_num	BYTE *	[in]	Indicates the number of infrared
	irCoverBitMask	BYTE *	[out]	The infrared occlusion status bit mask, such as 0b000000 1 (binary), is set to 1 in the 0th bit, indicating that the first infrared is occluded. 0b000000 11 (binary), is set to 1 in both the 0th and 1st bits, indicating that both the first and second infrared are occluded.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.5. LSG428 Special interface

5.5.1. LSG_FetchStoredDataAndAlarmFromGate

Definition	err_t LSG_FetchStoredDataAndAlarmFromGate(RFID_READER_HANDLE hr ,BOOLEAN del, DWORD additional,BYTE gid ,DWORD *pcInTotal,DWORD *pcOutTotal,DWORD * pcInDay,DWORD *pcOutDay) ;					
Description	Obtain the security door storage record and alarm					
Parameter	Name	Type	Direction	Remark		
	hr	RFID_READER_HANDLE	[in]	Handle of device driver		
	del	BOOLEAN	[in]	FALSE: Refers to obtaining the record pointed by the current record cursor; TRUE: Indicates that the current record cursor is first moved to a new position (i.e. current position+number of successfully uploaded records) and the record of new position is uploaded;		
	additional	DWORD	[in]	Extend option;		
	gid	BYTE	[in]	Target door ID of the alarm, only effective when additional.ALA=1 0: indicates that all security doors alarm; 1: Indicates the first security door alarm; 2: Indicates the second security door alarm;		
	pcInTotal	DWORD *	[out]	Total number of people entrance the library	Only effective when additional.PC_T=1	
	pcOutTotal	DWORD *	[out]	Total number of people leaving the library		
	pcInDay	DWORD *	[out]	Total number of people entrance the library on the same day	Only effective when additional.PC_DR=1	
	pcOutDay	DWORD *	[out]	Total number of people leaving the library on the same day		
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.					
Reference Code						

additional: extend option;

Position	7	6	5	4	3	2	1	0
Content	0	0	0	PC_DR	ALA	PC_R	PC_D	PC_T

PC_T: Take the total number of people in and out of the library;

PC_D: Obtain daily the current total number of people in and out records;

PC_R: Take the real-time change record of the day;

ALA: Enable the alarm. This position 1 indicates that the security door alarm is triggered while taking the record;

PC_DR: Take the current total number of people in and out on that day;

5.5.2. LSG_GetBufRecordCount

Definition	DWORD LSG_GetBufRecordCount(RFID_READER_HANDLE hr,BYTE type)			
Description	Get the number of traffic records read			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	type	BYTE	[in]	Record type: 0: Number of alarm records;, 1: Number of daily traffic records; 2: Number of real-time changes in daily traffic records;
Return (INT)	Return Number of alarm records;			
Reference Code				

5.5.3. LSG_ReadBufRecord

Definition	RFID_DN_HANDLE LSG_ReadBufRecord(RFID_READER_HANDLE hr , BYTE seek,BYTE type)				
Description	Read buffer record				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	seek	BYTE	[in]	Tag data report location	
				Value	Description
				0	The cursor position remains unchanged, and takes out the current record.
				1	Position the cursor to the first item and take out the current record.
				2	cursor locate the next , and take out current record.
				3	cursor locate the last , and take out current record.
	type	BYTE	[in]	Record type: 0: alarm record. 1: Passenger flow record. 2: real-time passenger flow record.	
Return (RFID_DN_HANDLE)	Returns the record data node specified by the current cursor				
Reference Code					

5.5.4. LSG_ParseAlarmRecord

Definition	err_t LSG_ParseAlarmRecord(RFID_DN_HANDLE hTagReport, BYTE *sid, BYTE *date, BYTE *eventType, DWORD *dFla, BYTE *uid, DWORD *uSize, BYTE *user, DWORD *userSize);			
Description	Parse the alarm record data of the security door			
Parameter	Name	Type	Direction	Remark
	hTagReport	(RFID_DN_HANDLE)	[in]	Node handle for tag data information
	sid	BYTE*	[out]	The security door ID that generate a recorded
	date	BYTE [7]	[out]	The date and time when the record was generated;
	eventType	BYTE *	[out]	Alarm event type
	dFla	DWORD*	[out]	Data flag
	uid	BYTE *	[out]	Book tag serial number Notice: The uid returned by LSG428 is 9 bytes, and the last byte is DSFID. The uid returned by LSG618 is EPC code;
	uSize	DWORD *	[in,out]	In: The maximum number of bytes that can be written to the uid buffer; Out: The number of bytes written to the buffer uid;
	user	BYTE *	[out]	User area block data;
	userSize	DWORD *	[in,out]	In: The maximum number of bytes that can be written to the user buffer; Out: The number of bytes written to the user buffer;
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

dFla: data flag;

Position	7	6	5	4	3	2	1	0
Content	0	0	0	0	USER	UID	DIR	

DIR: Direction of in and out;

0: indicates that the direction cannot be determined;

1: indicates in;

2: indicates out;

UID: When this bit is 1, it indicates that the parameters 'UID' and 'nSIZE' are valid;

USER: When this bit is 1, it indicates that the parameters 'USER' and 'userSize' are valid;

5.5.5. LSG_ParseDayPCRecord

Definition	err_t LSG_ParseDayPCRecord(RFID_DN_HANDLE hTagReport,BYTE *date,DWORD *inPC,DWORD *outPC) ;			
Description	Parse the recorded data of in and out the library people count in everyday			
Parameter	Name	Type	Direction	Remark
	hTagReport	(RFID_DN_HANDLE	[in]	Node handle for tag data information
	date	BYTE [4]	[out]	Date of record generation;
	inPC	DWORD *	[out]	The total number of people entrance on this day;
	outPC	DWORD *	[out]	The total number of people leave on this day;
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.5.6. LSG_ParseRealtimePCRecord

Definition	err_t LSG_ParseRealtimePCRecord(RFID_DN_HANDLE hTagReport, BYTE *datetime, BYTE *dir, DWORD *PC);			
Description	Parse the recorded data of real-time changes of the people in and out the library			
Parameter	Name	Type	Direction	Remark
	hTagReport	(RFID_DN_HANDLE)	[in]	Node handle for tag data information
	datetime	BYTE [7]	[out]	Date of record generation;
	dir	BYTE *	[out]	Access direction: 0: mean in; 1: mean out;
	PC	DWORD *	[out]	The total number of people on this day;
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.5.7. LSG_InitPC

Definition	err_t LSG_InitPC(RFID_READER_HANDLE hr ,BYTE type,DWORD pcTotal,DWORD pcDay) ;			
Description	init traffic			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	type	BYTE	[in]	Cumulative traffic type 0: Total cumulative passenger flow; 1: cumulative passenger flow in the same day;
	pcIn	DWORD	[in]	People entrance
	pcOUT	DWORD	[in]	People leaving
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.5.8. LSG_Alaram

Definition	err_t LSG_Alaram(RFID_READER_HANDLE hr);			
Description	Security door alarm			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	sid	BYTE	[in]	Security door ID: 0:all security door alarm; 1:first security door alarm; 2:second security door alarm;
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.5.9. LSG_FetchPCFromGate

Definition	err_t LSG_FetchPCFromGate(RFID_READER_HANDLE hr, BYTE option, DWORD *pcInTotal, DWORD *pcOutTotal, DWORD *pcInDay, DWORD *pcOutDay) ;			
Description	Obtain cumulative pedestrian traffic from the security door separately			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	option	BYTE	[in]	Option: 0x01: Obtain the cumulative total number of people entrance and leaving the library; 0x02: Obtain the current number of people entrance and leaving the library on that day; 0x03: Obtain both of the above;
	pcInTotal	DWORD *	[out]	Total number of people entrance the library
	pcOutTotal	DWORD *	[out]	Total number of people leaving the library
	pcInDay	DWORD *	[out]	Total number of people entrance the library on the same day
	pcOutDay	DWORD *	[out]	Total number of people leaving the library on the same day
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

5.5.10. LSG_GetIRTriggerCounter

Definition	err_t RFIDLIB_API LSG_GetIRTriggerCounter(RFID_READER_HANDLE hr, BYTE sid, DWORD *ir_tri_cnt, BYTE *ir_num);			
Description	Obtain the times of infrared shielding of the security door			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	sid	BYTE	[in]	For 40X, default 0 For 42X and 61X series, according actual door number
	ir_tri_cnt	DWORD *	[in,out]	Number of infrared occlusions, such as ir_num is an array byte[] {1,2}, then ir_tri_cnt[0] is the number of occlusion times for infrared 1, ir_tri_cnt[1] is the number of occlusion times for infrared 2.
	Ir_num	BYTE *	[in]	Infrared number array, such as passing an array byte[] {1,2}, the infrared devices it needs to obtain the number of infrared occlusion times are infrared 1 and infrared 2
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

6. File_box_positioning API

6.1. DOC_SetLock

Definition	int DOC_SetLock(RFID_READER_HANDLE hr, BYTE lockID, BYTE lockSta)			
Description	Control lock switch			
Parameter	Name	Type	Direction	Remark
	Hr	RFID_READER_HANDLE	[in]	Device handle
	LockID	byte	[in]	Lock number
	LockSta	BYTE	[in]	Lock flag; 【0 close, 1 open】
Return (err_t)	0: Success Not 0: Failure			

6.2. DOC_GetLockSta

Definition	int DOC_GetLockSta(RFID_READER_HANDLE hr, BYTE lock, WORD* lockState)			
Description	Control lock switch			
Parameter	Name	Type	Direction	Remark
	Hr	RFID_READER_HANDLE	[in]	Device handle
	lock	BYTE	[in]	Lock enable bit. The corresponding bit represents the lock number. Such as 0x03:0000 0011, indicates the status of obtaining locks 1 and 2
	LockState	WORD*	[out]	Lock status. The corresponding bit represents the status of the corresponding lock number. The corresponding bit is 0 for close/alarm, and 1 for open/normal. Bit 1 and bit 2 represent the lock status and error status of lock 1 respectively. For example, 0x000B, low byte 0000 1100, bit 3 and bit 4 represent lock number 2. The third digit represents the lock status: 1 is open, the fourth digit represents the error status, and 1 is normal.
Return (err_t)	0: Success Not 0: Failure			

6.3. DOC_SetLight

Definition	int DOC_SetLight(RFID_READER_HANDLE hr,BYTE lightMode)			
Description	Light Control			
Parameter	Name	Type	Direction	Remark
	Hr	RFID_READER_HAN DLE	[in]	Device handle
	lightMode	BYTE	[in]	【0 close、1 open】
Return (err_t)	0: Success Not 0: Failure			

6.4. LED Control

6.4.1. DOC_CreateLightControlList

Definition	RFID_DN_HANDLE DOC_CreateLightControlList();			
Description	Create LED parameter list			
Parameter	Name	Type	Direction	Remark
Return (err_t)	Return LED handle, use for DOC_AddLightControl interface as paramter			

6.4.2. DOC_AddLightControl

Definition	Int DOC_AddLightControl(RFID_READER_HANDLE hr,RFID_DN_HANDLE hTagsSet, BYTE side, BYTE layer, WORD column, WORD mask, BYTE color, BYTE Mode, DWORD val, WORD flash_AT, WORD flash_DT)				
Description	Add light parameter list				
Parameter	Name		Type	Direction	Remark
	hr		RFID_READER_HANDLE	[in]	Device handle
	hLightSet		RFID_DN_HANDLE	[in]	LED handle create by DOC_CreateLightParamList
	buff	side	BYTE	[in]	【1 A side, 2 B side】
		layer	BYTE		Layer number
		column	WORD		Set 0, reserved
		mask	WORD		Column number, where the corresponding bit means correspond column.
		Color	BYTE		Light color: 【0 red、1 green】
		mode	BYTE		Light mode: 【0 turn off、1 light always、2 flash always、3 light N ms、4 flash N ms、5 flash N times】
		val	DWORD		When MODE=0x03, it indicates the time when the light is on When MODE=0x04, it indicates how long it flashes; When MODE=0x05, it indicates how many times it flashes;
		Flash_AT	WORD		Active time
		Flash_DT	WORD		Unactive time
Return (err_t)	0: Success Not 0: Failure				

6.4.3. DOC_LightControl

Definition	int DOC_LightControl(RFID_READER_HANDLE hr, RFID_DN_HANDLE hLightSet);			
Description	LED control			
Parameter	Name	Type	Direction	Remark
	Hr	RFID_READER_HANDLE	[in]	Device handle
	hLightSet	RFID_DN_HANDLE	[in]	LED parameter list, Create by DOC_CreateLightControlList
Return (err_t)	0: Success Not 0: Failure			

6.5. DOC_GetLayerReaderInfor

Definition	int DOC_GetLayerReaderInfor(RFID_READER_HANDLE hr, BYTE Type, BYTE side, BYTE layer, LPTSTR buffer,DWORD *nSize);			
Description	Get reader information			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Device handle
	type	BYTE	[in]	Set 1, means reader
	Side	byte	[in]	Which side, The cabinet is divided into A side and B side 【1 A side, 2 B side】
	Layer	byte	[in]	Layer number
	Buffer	LPTSTR	[in,out]	Layer Reader Information
	nSize	DWORD*	[in,out]	Buffer field size
Return (err_t)	0: Success Not 0: Failure			

6.6. DOC_SystemErrorFlag

Definition	int DOC_SystemErrorFlag(RFID_READER_HANDLE hr, DWORD* flag);			
Description	Get system error flag			
Parameter	Name	Type	Directory	remark
	Hr	RFID_READER_HANDLE	[in]	Device handle
	Flag	DWORD*	[in,out]	System error flag
Return (err_t)	0: Success Not 0: Failure			

6.7. Tag Inventory

6.7.1. DOC_StartInventory

Definition	int DOC_StartInventory(RFID_READER_HANDLE hr, BYTE layer, BYTE column, BYTE readMode, WORD offset, BYTE cnt);			
Description	Start inventory			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Device Handle
	Layer	Byte	[in]	Layer. Default inventory A and B side, if only inventory A side, need to OR-operation with 0x40, B side: OR-operation with 0x80. E.g. 0x41:it means first layer in A side
	Column	Byte	[in]	column, default 0
	ReadMode	Byte	[in]	0:UID and USER; 1:UID;
	Offset	WORD	[in]	Datablock offset
	Cnt	Byte	[in]	Count of datablock
Return (err_t)	0: Success Not 0: Failure			

6.7.2. DOC_GetInventoryReport

Definition	int DOC_GetInventoryReport(RFID_READER_HANDLE hr, BYTE flash);			
Description	Extract tag data report from device			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Device handle
	Flash	Byte	[in]	【 0 get previous fail record 、 1 get current record 】 , set flash=1 when inventory , if fail then set flash =0 to retry again
Return (err_t)	0: Success Not 0: Failure			

6.7.3. DOC_GetTagDataReport

Definition	RFID_DN_HANDLE DOC_GetTagDataReport(RFID_READER_HANDLE hr, BYTE seek);				
Description	Get tag data report				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Device handle	
	seek	BYTE	[in]	The tag data report position	
				Value	Description
				0	The cursor position remains unchanged, get current record。
				1	The cursor move to first, get current record
				2	The cursor move to next, get current record
	3	The cursor move to last, get current record			
Return (RFID_DN_HANDLE)	Return data node of tag data report that specified by current cursor.				

6.7.4. DOC_ParseTagDataReport

Definition	int DOC_ParseTagDataReport(RFID_READER_HANDLE hr, RFID_DN_HANDLE hTagReport, BYTE* layer, BYTE* column, BYTE* uid_len, BYTE uid[], BYTE* db_len, BYTE db_data[]);			
Description	Parse tag data report			
Parameter	Name	Type	Direction	Remark
	Hr	RFID_READER_HANDLE	[in]	handler
	hTagReport	RFID_DN_HANDLE	[in]	Tag handle
	Layer	Byte*	[in,out]	Side and layer.front 2 bit means layer number, 0x80:B side, 0x40:A side, behind 6 bit means layer
	column	Byte*	[in,out]	column
	uid_len	Byte*	[in,out]	Length of UID
	Uid	Byte[]	[in,out]	UID
	db_len	Byte*	[in,out]	Length of datablock
	db_data	Byte[]	[in,out]	User data
return (err_t)	0: Success Not 0: Failure			

AppendixA. Event Type Description

Event type	Value	Description	
RFIDLIB_EVNT_GETBUF_REPORT_FOUND	1	Automatic acquisition function, report the output event	
		Parameter	Description
		wParam	Handle of device driver
		lParam	Reserved
RFIDLIB_EVNT_GETBUF_ERROR	2	Automatic acquisition function, the error output event	
		Parameter	Description
		wParam	Handle of device driver
		lParam	Error code
RFIDLIB_EVNT_GETBUF_EXIT	3	Automatic acquisition function, thread exit event	
		Parameter	Description
		wParam	Handle of device driver
		lParam	Reserved